

Difference Between JPA, Hibernate and Spring Data JPA(Handson-4)

Java applications often need to interact with databases. JPA, Hibernate, and Spring Data JPA help developers perform database operations more easily by reducing the amount of SQL and JDBC code.

1. JPA (Java Persistence API)

JPA is a **Java specification** that defines a standard way to map Java objects to database tables. It only provides interfaces and guidelines, so it cannot perform database operations by itself. JPA requires an implementation such as Hibernate.

- It is a specification (not a framework).
- Defines standard APIs for object-relational mapping (ORM).
- Needs an implementation like Hibernate.

2. Hibernate

Hibernate is an **ORM framework** and one of the most popular implementations of JPA. It provides the actual functionality to connect Java applications with databases and performs CRUD operations automatically.

- Implements JPA.
- Converts Java objects into database records.
- Reduces SQL and JDBC code.
- Supports advanced features like caching and lazy loading.

3. Spring Data JPA

Spring Data JPA is a Spring module built on top of JPA. It further simplifies database access by providing ready-made repository interfaces. Developers can perform most database operations without writing implementation code.

- Built on top of JPA.
- Uses repositories such as **JpaRepository**.
- Automatically provides methods like **save()**, **findAll()**, **findById()**, and **delete()**.

Comparison :

Feature	JPA	Hibernate	Spring Data JPA
Type	Specification	ORM Framework	Spring Module
Implementation	No	Yes	Uses JPA
Performs Database Operations	No	Yes	Yes
CRUD Support	Through implementation	Yes	Yes (using repositories)
SQL Code Required	Less	Very Less	Minimal

Conclusion

JPA defines how database operations should be done, Hibernate performs those operations, and Spring Data JPA makes the process simpler by providing ready-made methods. Together, they make Java database development easier and save development time.